

COMPUTER PROGRAMMING

LAB REPORT 1

Installation of DEV-C++ IDE and Running the First C++ Program

1. OBJECTIVE

- The purpose of this lab is to understand the installation and use of the DEV-C++ Integrated Development Environment (IDE) and to execute a basic C++ program successfully. Students also learn how to upload their work to GitHub for submission and version control.

2. THEORY

- C++ is a high-level programming language widely used in software development and engineering applications. To write and run C++ programs, programmers use an Integrated Development Environment (IDE). DEV-C++ is a beginner-friendly IDE that provides tools such as a code editor, compiler, and output console in one environment. Every C++ program starts from the `main()` function, which acts as the entry point of execution. The `cout` statement is used to display output on the screen.

3. SOFTWARE

- DEV-C++ IDE
- C++ Compiler (included with DEV-C++)
- GitHub Account

4. PROCEDURE / PROCESS

Step 1: Install DEV-C++

1. Download DEV-C++ from its official website.
2. Install the software on the computer.
3. Launch DEV-C++ after installation.

Step 2: Create a New C++ Program

4. Click File → New → Source File.
5. A new blank editor window will appear.

6. Click File → Save As.
7. Save the file with the name lab1_hello.cpp.

Step 3: Write the First C++ Program

```
#include <iostream>
using namespace std;

int main()
{
    cout << "Hello World!";
    return 0;
}
```

Step 4: Compile and Run the Program

8. Click Execute → Compile.
9. Then click Execute → Run.
10. The output window will display the result of the program.

5. GITHUB TASK

1. Open the official GitHub website: GitHub
2. Click on the Sign Up button.
3. Enter your email address, username, and password.
4. Verify your account and complete the registration process.
5. After successful registration, log in to your GitHub account.

Step 2: Create a Repository Named “Cpp-Lab-Tasks”

1. After logging in, click the + icon in the top-right corner.
2. Select New repository.
3. In the repository name field, type Cpp – lab - Tasks.
4. Choose the repository visibility as Public .
5. Click the Create Repository button.

Step 3: Upload the File “lab1_hello.cpp”

1. Open the created repository Cpp-Lab-Tasks.
2. Click on the Add File button.
3. Select Upload Files.

4. Drag and drop the file lab1_hello.cpp or click Choose Your Files to browse the file from the computer.
5. After uploading, click the Commit Changes button.

GitHub Repository Link: https://github.com/Zaid1477/Computer_programming

- The GitHub repository stores the lab files securely online and helps in maintaining backup, version control, and easy sharing of programming tasks.

6. CONCLUSION

- In this lab session, we learned how to install and use the DEV-C++ IDE to create, compile, and run a basic C++ program successfully. We also learned how to use GitHub for uploading and managing programming files. This lab improved our understanding of programming tools and basic software development practices.